SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I JITESH AP (192511163) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jitesh

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks 32 **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 - AT1

ANALYTICAL PROBLEM SOLVING

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(1) Given :

Ip address block : 192.168.20.0/24

* lab A subnet : /26
* lab B subnet : /27
* lab C subnet : /28

Step 1 : lab A (/26)

* prefix length : /26
* subnet mask : 255.255.255.192
* Total Ip address = 2^(32-26) = 64
* usable Host Address = 64 - 2 = 62

Details :

* Network Address : 192.168.20.64
* first Host Address : 192.168.20.65
* last Host Address : 192.168.20.62
* Broadcast Address : 192.168.20.63

Step 2 : Lab B (/27)

* prefix length : /27
* subnet mask : 255.255.255.224

* Total Ip addresses = 2^(32 - 27) = 32
* usable Host Addresses = 32 - 2 = 30

Details:

* Network Address : 192.168.20.64
* first Host address : 192.168.20.65
* Last Host address : 192.168.20.94
* Broadcast Address : 192.168.20.95

Step 3: lab C (/28)

Prefix length : /28
Subnet mask : 255.255.255.240
Total Ip Addresses = 2^(32 - 28) = 16
usable Host Addresses = 16 - 2 = 14

Details:

Network address : 192.168.20.96

First Host Address : 192.168.20.97

Last Host Address : 192.168.20.110

Broadcast Address : 192.168.20.111

Given:

System 1 Ip address : 192.168.1.130

system 2 Ip address : 192.168.1.190

Subnet Mask : 255.255.255.192

CIDR Notation : /26

Total Ip addresses = 2^(32 - prefix len)

usable Host address = 2^(32 - prefix len) - 2

for /26 :

Total Ip address $= 2^6 = 64$

usable Host address $= 64 - 2 = 62$

Step 1: subnet range

192.168.1.0 – 192.168.1.63

192.168.1.64 – 192.168.1.127

192.168.1.128 – 192.168.1.191

192.168.1.192 – 192.168.1.255

Step 2: Network Address

The network address of the subnet is

192.168.1.128

Step 3: Broadcast Address

The broadcast address is the last Ip address in the subnet : 192.168.1.191

Step 4: Valid Host Range

* First Host address : 192.168.1.129
* Last Host address : 192.168.1.190

Valid host range is:

192.168.1.129 - 192.168.1.190

Step 5: Verify given Ip address

Sys 1:

* Ip Address : 192.168.1.130
* falls within the valid host range
* Therefore, system 1 belongs to subnet 192.168.1.128 /26

Sys 2:

Ip address : 192.168.1.190

falls within the valid host range

Therfore, system 2 also belong to subnet 192.168.1.128 /26

Step 1: HR Department (/25)

* Subnet Mask : 255.255.255.128
* Total Ip Address = 128
* usable Host = 126

Details:

* Network Address : 172.16.0.0
* First Host : 172.16.0.1
* Last Host : 172.16.0.12.6
* Broadcast Address : 172.16.0.127

Step 2: Sales Department (/26)

* Subnet Mask : 255.255.255.192
* Total Ip address = 64
* usable Host = 62

Details:

* Network Address : 172.16.0.128
* First Host : 172.16.0.129
* Last host : 172.16.0.190
* Broadcast : 172.16.0.191

Step 3: IT Department (/27)

* Subnet Mask : 255.255.255.224
* Total Ip address = 32
* usable Host = 30

Details:

* Network address : 172.16.0.192
* First Host : 172.16.0.193
* Last Host : 172.16.0.222
* Broadcast : 172.16.0.223

Step 4: Admin Department (/28)

Subnet mask : 255.255.255.240

Total IP : 16

Usable Host : 14

Details:

* Network Address : 172.16.0.224
* first Host : 172.16.0.255
* Last Host : 172.16.0.238
* Broadcast : 172.16.0.239

Step 5: Remaining unused ip address range

The remaining unused ip address are:

172.16.0.240 - 172.16.255.255

Ste

ep 1: Calculate the number of students

Original prefix = /24

New prefix = /27

No. of subnets = 2^(27-24) = 2^3 = 8

Total no. of subnet = 8

Step 2: calculate no. of usable Hosts

* Host bits = 32 - 27 = 5

Total Ip address = 2^5 = 32

Usable host address = 32 - 2 = 30

usable host per subnet = 30

Step 3: subnet Table

Subnet	Network Address	Broadcast address
1	172.16.10.0	172.16.10.31
2	172.16.10.32	172.16.10.63
3	172.16.10.64	172.16.10.95
4	172.16.10.96	172.16.10.127
5	172.16.10.128	172.16.10.159
6.	172.16.10.160	172.16.10.191
7.	172.16.10.192	172.16.10.223
8.	172.16.10.224	172.16.10.255

Step 1: Compress the IPV6 Address

rules:

* Remove leading zeros in each block
* Replace consecutive block of zeros with ::

compressed IPV6 Address:

2001:db8::ff00:42:8329

Step 2: Expand the IPV6 Address

Expand the compressed address by restoring all omitted zeros

Expanded IPV6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine network prefix

A /64 prefix means the first 64 bits represent Network portion.

Network prefix:

2001:0db8:0000:0000::/64